

NUMERICAL EXPERIMENTAL ANALYSIS OF FILTERING MODEL SCALING PROCEDURE

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Abstract. In this paper, a near-wellbore distance uplift procedure is proposed for finite-permeability hydraulic fractures on a Voronoi network, and it is shown that fracture cells are clearly present on a coarse grid. Embedded discrete fracture model, it is shown that the assumption of linearity of pressure distribution in a fractured reservoir cell can lead to a significant error in solving the global problem. In this, the accuracy of solving the problem by changing the permeability of the cells was significantly analyzed.

Key words: procedure, shortened, component, porosity, hydraulic, isotropic conductivity field, hydraulic and reservoir.

Modeling the behavior of a three-phase mixture in a reservoir and in fractures consists in solving the following problem, which is called global.

$$\frac{\partial}{\partial t} (\varphi \rho_{\alpha} S_{\alpha}) + \nabla \cdot (\rho_{\alpha} \vec{v}_{\alpha}) = \rho_{\alpha} q_{\alpha} \delta_{well} \quad (1.1)$$

$$\vec{v}_{\alpha} = - \frac{k k_{r\alpha}}{\mu_{\alpha}} \nabla p_{\alpha}, \quad (1.2)$$

$$\sum_{\alpha} S_{\alpha} = 1, \quad (1.3)$$

$$\left. \frac{\partial \rho_{\alpha}}{\partial n} \right|_{\partial \Omega} = 0 \quad (1.4)$$

here, α - component index (water, oil or gas); φ - porosity of rocks; ρ_{α} - density; S_{α} - saturation; $\vec{v}_{\alpha}$ - speed of the component; k - isotropic conductivity field; $k_{r\alpha}$ - relative phase conductivity; μ_{α} - viscosity; p_{α} - pressure; q_{α} - flow of components from the well; δ_{well} - Dirac delta function (dimensions 1/m³), localized at the location of the injection well; $\partial \Omega$ - the outer boundary of the considered domain Ω . In this case, it is possible to set different boundary conditions at the well sites and different initial conditions throughout the field. Area Ω includes hydraulic fractures. That is, the behavior of the three-phase mixture of water in cracks is described by the same equations.

Spatial discretization of equations (1.1-1.4) in the finite volume method using the two-point flow approximation and transformation of the derivative leads to a system of nonlinear equations whose left-hand finite difference over time is usually solved by Newton's method. In this case, the fluid flow between the two cells is calculated as follows:

$$q_{\alpha,ij} = -T_{ij} \frac{k_{r\alpha}}{\mu_{\alpha}} (\rho_{\alpha,i} - \rho_{\alpha,j}),$$

here, the single index i -value in cell, paired index- ij , and j indicates the value at the edge between cells. If cells i and j assuming that the pressure between the nodes is linearly distributed, then we can use. The linear formula for calculating the conductivity between these cells is:

$$T_{i,j} = k_{i,j} \frac{A_{i,j}}{d_{i,j}},$$

here, $A_{i,j}$ - the area of the face between the cells i and j , $d_{i,j}$ -distance between nodes i and j . To calculate $k_{i,j}$ - common harmonic mean k_i and k_j . However, cells i and j can represent both

layer control volumes and fracture control volumes. r is the well field because it is a solution to Poisson's equation:

$$\rho(r) \sim \ln(r).$$

In this regard, the use of a linear formula for the calculation of conductivity in the area near the well will introduce some errors in the calculation of flows. Lifting method $T_{i,j}$ is shortened to reflect the non-linear nature of the pressure distribution between the cells by the coefficient.

Scaling procedure. For each well, the auxiliary local problem is solved on a fine-grained mesh:

$$\begin{aligned} \nabla \cdot (k \nabla \rho) &= 0, \\ \rho|_{\partial\psi} &= 0 \\ \rho|_{\omega} &= 1. \end{aligned}$$

here, $PS \subset \Omega$ is the area near the well, including hydraulic fracturing, $\partial\psi$ — external border of the region and PS , $\omega \in \Omega \subset \Omega$ is the center of the well. In this approach, hydraulic fracturing is modeled by creating a specific region k with increased permeability. In this, the method of spatial discretization of the problem plays a special role: the region Ω is divided in such a way that the boundaries of the partition elements correspond to the boundaries of the field of enhanced conductivity.

It should be noted that in some of the same problems, p can be expressed analytically, but in this work, it is assumed that the water has heterogeneous isotropic characteristics, so a numerical solution is used. The solution contains detailed information about the nonlinear nature of the pressure distribution in the p region at steady state. For the global problem, the permeability between large cells is calculated by solving. In which p is as follows:

$$T_{i,j} = -\frac{q_{i,j}}{\langle \rho \rangle_i - \langle \rho \rangle_j} = -\frac{\int_{V_i \cap V_j} k \nabla p \cdot d\vec{s}}{\frac{1}{|V_i|} \int_{V_i} p dv - \frac{1}{|V_j|} \int_{V_j} p dv}, \quad (1.5)$$

here, $q_{i,j}$ — is the single-phase fluid flow between cells i and j ; $\langle \rho \rangle_i$ — average pressure value p in cell i , V_i i - cell, $|V_i|$ — volume i - cell, $V_i \cap V_j$ - the border between i and j cells.

The algorithm for solving the global problem using the scaling procedure consists of four steps:

1. Construction of global and local computing networks for regions Ω and PS , respectively.
2. To solve the local problem described above in the area of PS .
3. Calculate the throughput for the global computing network in the region PS according to the formula described above and in the region $\Omega \setminus PS$ according to the linear formula in Section 2.
4. Finding a solution to the global problem in the region, using the obtained permeability - Ω .

Mesh construction. It is proposed to construct two coherent computing networks for solving global and local problems: a coarse network for the global problem and a fine network for local problems. Coarse and fine meshes are consistent if:

- each coarse grid cell can be described as a union of several fine grid cells;
- each face of a coarse mesh can be represented as a union of several fine mesh faces.

The consistency property allows accurate calculation of the values of the integrals over the cells and faces of the coarse mesh to calculate the conductivity. You can use the following algorithm to construct consistent unstructured meshes.

1. Create a mesh for the entire area Ω , where the areas close to the well are detailed with small cells, and the watery areas far from the wells are detailed with large cells. This network is called a fine grid and is used to solve local problems.

2. Construction of a new grid for the entire area based on the fine grid By combining small cells near the Ω well into large cells, the new grid is called a coarse grid and is used to solve the global problem.

In the first step, a fine mesh is constructed. In the near-wellbore region, nodal points are placed in increasing steps along the hydraulic fracture normal and radially at the fracture tips, as shown in Figure 1.1. Wells with hydraulic fractures on a fine grid are represented by a dimensional fact model (DFM). In this case, the control volumes for refraction are built on the surfaces between the water control volumes (Fig. 1.2). Fracture cells are represented geometrically as one-dimensional lines. But having a non-zero volume is calculated from the width of the fracture opening. To ensure that the total volume of the near-well area does not change, the volume of adjacent reservoir cells adjacent to the fracture is reduced accordingly, and a fine grid is constructed with fractured adjacent reservoir cells.

In the second step, a coarse mesh is constructed. The fine-mesh reservoir cells are combined into coarse meshes as shown in Figure 1.3. In this case, the crack cells will not change and will also exist in the coarse mesh. But each fracture cell is connected to only one coarse reservoir cell.



Location of nodal points for constructing a fine computational grid near a hydraulic fracture Fig. 1.1.

Node points are shown in gray. The black line indicates hydraulic fracturing.

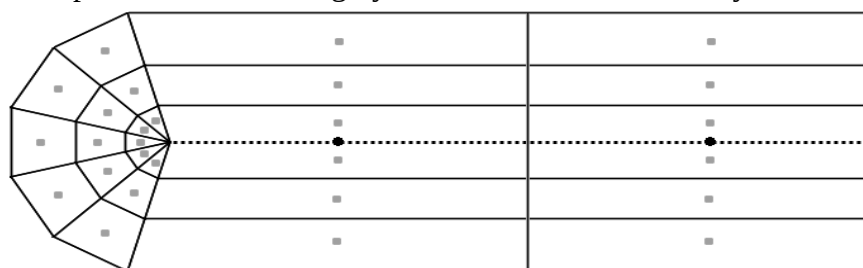


Figure 1.2. A section of a fine calculation grid near a hydraulic fracture. Nodal reservoir cell points are shown in gray. Cell attachment points are indicated by cracks in black.

The black line indicates hydraulic fracturing, water intrusion into production wells, or oil degassing when starting new wells and is much shorter than the entire simulation period. The product of the conductance and area of the broken cells is comparable to the product of the conductance and area of the reservoir cells, so the broken cells should not have a strong negative effect on the conditionality of the system of linear equations.

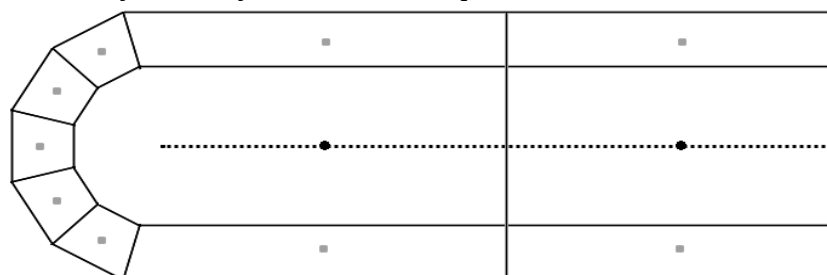


Figure 1.3. Part of a rough calculation grid near a hydraulic fracture.

All signs are similar to those in the picture. The dotted line represents only broken cells, not the line between cells. The nodal points of the rough cells through which the crack passes are geometrically compatible with the nodal points of the crack cells.

A digital experience. As an example, the problem of two-phase filtration in an inhomogeneous isotropic reservoir is considered. In the area under consideration, there are production and injection wells with hydraulic fracturing operating under constant bottom pressure. The opening width of both cracks is 1mm. The production well fracture has a permeability of 500,000 Darcy to simulate an infinite permeability, and the injection fracture has a permeability of 500 Darcy. Figure 1.6 shows the pressure obtained by solving two local problems on a fine mesh. This pressure is used to show the conductivity calculations in order of increasing scale. It can be seen that the pressure distribution in the two local areas is different due to the different fracture permeability.

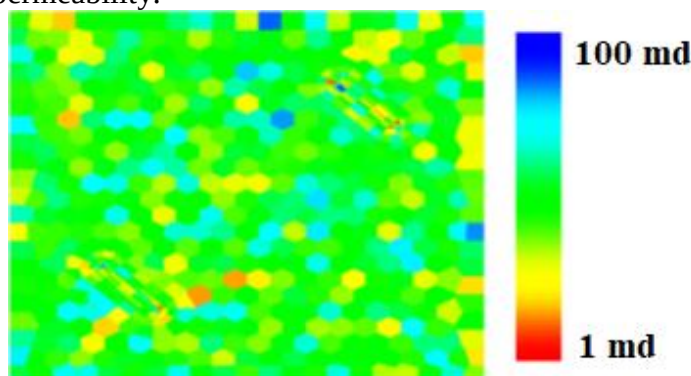


Figure 1.4. Reservoir permeability.

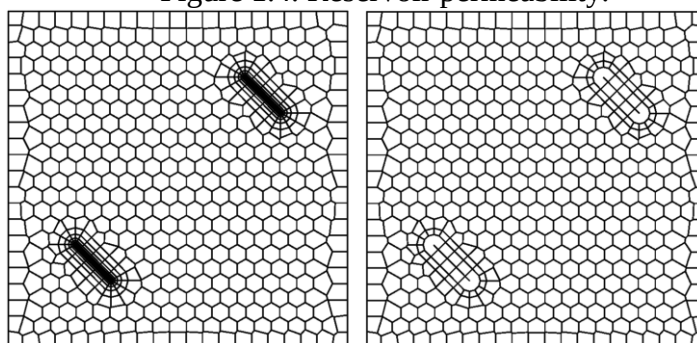


Figure 1.5 - Fine mesh (left) and coarse mesh (right).

In the lower left corner is the extraction well, and in the upper right corner is the injection well. 3 options for calculating the global problem were conducted. The first calculation is carried out on a fine grid using the classical linear formula to calculate all the conductances between cells and taken as a reference. The second calculation is performed on a coarse grid using a scaling procedure.

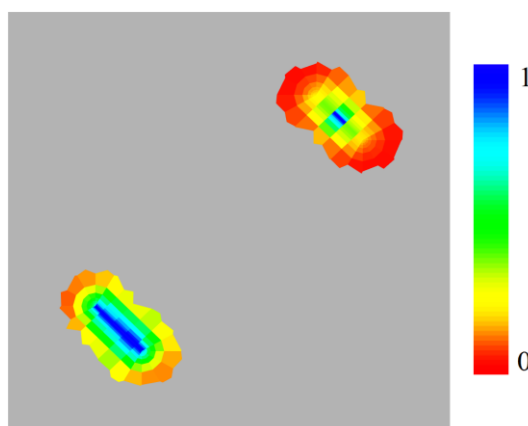


Figure 1.6. Pressure obtained by solving two local problems on a fine mesh.

The third calculation is performed on a coarse grid using the classical linear formulation and is equivalent to using an embedded discrete fracture model (EDFM). In this case, the permeability between the rough reservoir cell i and the crack cell j is calculated according to the formula.

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